ZIJIE (FRANK) GUO

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EDUCATION & CERTIFICATIONS

UNIVERSITY OF CHICAGO — Chicago, IL
Master of Science in Applied Data Science, 2-Year Thesis Track — Expected Jun 2028
- **Merit-Based Tuition Scholarship Recipient** ($15,000)
- **Relevant Coursework:** Statistical Models for Data Science; Cloud-Native Data Engineering

GEORGE WASHINGTON UNIVERSITY — Washington, DC
Bachelor of Science in Business Analytics, Concentration in Finance — May 2026
- **GPA:** 3.74/4.00 | Magna Cum Laude | Dean's List
- **Relevant Coursework:** Machine Learning; Forecasting Analytics (Time Series Analysis)

CFA INSTITUTE | Passed Level I of the CFA Program — Aug 2026

PROFESSIONAL EXPERIENCE

Dongxing Securities — Shanghai, China
Quantitative Research Intern, CTA Desk — Jul 2025 - Aug 2025
- Analyzed 80+ alpha factors across Chinese equity index futures (IF, IC, IM) using price, volume, order-flow, and market-depth data; built a Python XGBoost framework with 5-fold rolling validation, ranking factors by mean feature importance and cross-fold stability to select the top 30.
- Designed a lag-only feature pipeline with per-window transforms to preserve time alignment and prevent look-ahead bias; separated structural and technical missingness with a NaN-preserving scheme and is_active flag.
- Implemented a sliding-window out-of-sample validation and net-of-cost backtest workflow (12-month training, 3-month validation) to test whether factor signals generalized beyond the training period.
- Delivered annotated Python scripts, Jupyter notebooks, dashboards, and technical documentation.

Financial Street Securities (formerly Hengtai Securities) — Suzhou, China
Summer Fixed Income Analyst — Jul 2023 - Aug 2023
- Conducted due diligence on 30+ enterprises for urban investment bond issuance using the National Credit Information System and other regulatory platforms, supporting the underwriting of ¥1.45B in bonds.

PROJECTS

Autonomous Factor Research Harness | Microsoft Qlib — Feb 2026 - Present
- Adapted Karpathy's autoresearch loop to quantitative factor research: an LLM agent proposed factor expressions while a fixed, agent-immutable evaluator enforced time-separated discovery (2021), validation (2022), and holdout (2023) gates plus correlation de-duplication.
- Ran 4,622 cloud trials across 19 signal families; after finding a documented holdout gate was never enforced in code, added schema-level validation that removed 197 sign-flipped factors (30% of the archive).
- Attributed the lead strategy's performance and found a significant cash/beta contribution rather than pure alpha, repositioning the project from alpha hunting to a reproducible research platform.

Procurement Pilot | System Architect — Jan 2026 - May 2026
- Led architecture and integration for a four-person capstone, unifying independently developed analytics and optimization modules into a Streamlit decision workflow; the project earned the Faculty Choice Award, and the team presented the system at FI Consulting's office.
- Re-architected a keyword-routed prototype into a hybrid LangGraph system: an LLM handled open-ended intent and task planning, while deterministic parsers and direct analytics tools executed decision-critical workflows, reducing recurring query execution from ~120 seconds to sub-second latency.
- Designed pre-execution request review and integrated post-solve approve/modify/discard controls; exposed agent selection, parameters, outputs, and latency through execution traces to keep recommendations auditable.

ACTIVITIES

- Member, Delta Sigma Pi Professional Business Fraternity, Beta Alpha Chapter
- Event Organizer & Sponsor Coordinator, GWCCA - organized events for 70+ attendees and secured ~$2,000 in sponsorships

SKILLS & INTERESTS

- **Programming & Research Tools:** Python (pandas, NumPy, SciPy, scikit-learn, XGBoost), SQL (MySQL), R, Bash, Git
- **Cloud & Data Platforms:** Google Cloud (Cloud SQL), Snowflake, Databricks (Spark, Delta Lake), REST APIs
- **Quantitative Research:** Factor research, feature engineering, rolling-window validation, backtesting, and leakage prevention
- **AI Engineering:** Evaluation harness design, agent research loops, schema-enforced validation, human-in-the-loop review
- **Languages & Interests:** English (fluent), Mandarin (native); investment research, basketball, and weightlifting